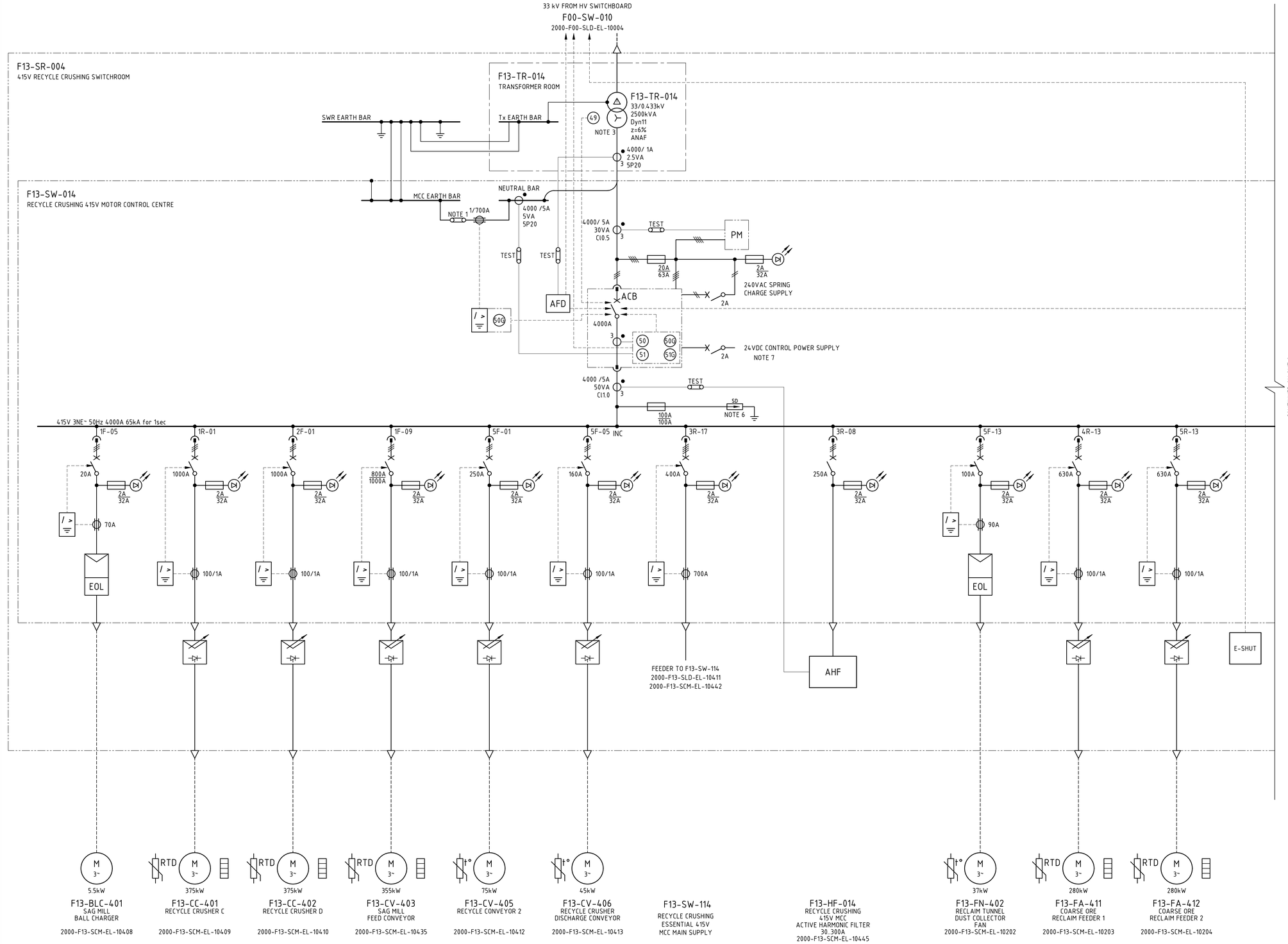


NOTES

1. EARTH-NEUTRAL LINK SHALL BE INSTALLED IN INCOMER TIER. SIGNAGE SHALL BE PLACED ON MCC AND ON TRANSFORMER READING "M.E.N. LINK LOCATED IN F13-SW-014 INCOMER TIER" CABLE SHALL BE IDENTIFIED C/W LABEL STATING "M.E.N. LINK - DO NOT REMOVE".
2. SWITCHBOARD SHALL BE EXTENSIBLE AND BE PROVIDED WITH THE FACILITY TO EXTEND THE MAIN BUSBAR AND APPEND ADDITIONAL TIERS AT ONE END WITHOUT MAJOR STRUCTURAL DISASSEMBLY.
3. UPSTREAM TRANSFORMER FEEDER SHALL BE TRIPPED ON ACTIVATION OF SWITCHROOM EMERGENCY PUSHBUTTON, DETECTION OF EARTH FAULT OR ARC FAULT DETECTION.
4. SWITCHBOARD VENDOR SHALL SELECT SWITCHGEAR TO PROVIDE TYPE-2 CO-ORDINATION SUITABLE FOR HIGH EFFICIENCY MOTORS.
5. ALL DRIVE I/O VIA COMMS NETWORK TO DCS. FOR COMMUNICATIONS BLOCK DIAGRAM REFER 2000-F13-BLK-IC-10401.
6. SURGE DIVERTERS SHALL PROVIDE FEEDBACK TO DCS VIA VOLT-FREE CONTACTS. SURGE DIVERTER FUSES SHALL BE SIZED BY VENDOR TO SUIT DUTY.
7. MINIATURE CIRCUIT BREAKERS SHALL BE SIZED AND RATED TO SUIT ACB AND INCOMER REQUIREMENTS.
8. THERMAL OVERLOAD UNITS MAY BE SUBSTITUTED WITH ELECTRONIC (MAGNETIC) OVERLOAD UNITS TO SUIT DUTY.
9. TOLs TO BE MOUNTED IN PURPOSE-BUILT ENCLOSURE BENEATH VSD. TOLs MAY BE INCLUDED IN MCC MODULE WITH ENGINEERING APPROVAL.
10. ANY DRIVE RTDs SHALL BE WIRED TO DEDICATED REMOTE I/O CARDS IN THE FIELD. RTD RELAYS MAY BE INCLUDED IN THE MCC MODULE WITH ENGINEERING APPROVAL.
11. CIRCUIT BREAKER TO BE SUBSTITUED FOR FUSED-SWITCH WHERE REQUIRED BY EQUIPMENT MANUFACTURERS GUIDELINES.
12. EMERGENCY SHUTDOWN (E-SHUT) PUSH BUTTON N/O WIRED TO SHUNT TRIP NOMINATED CIRCUIT BREAKERS.

LEGEND

- DIRECT ON-LINE MOTOR STARTER GENERIC - NOTE 4, 5
- DIRECT ON-LINE MOTOR STARTER REVERSING - NOTE 4, 5
- EOL — ELECTRONIC OVERLOAD UNIT NOTE 4, 5
- THERMAL OVERLOAD UNIT NOTE 8
- VARIABLE SPEED DRIVE NOTE 4, 5
- EARTH LEAKAGE RELAY NOTE 4, 5
- FIELD-MOUNTED DECONTACTOR
- MOTOR THERMISTOR (ONLY REQ. FOR MOTORS >37kW - 250kW)
- RTD — MOTOR RTD (ONLY REQ. FOR MOTORS >250kW)
- MOTOR HEATERS (ONLY REQ. FOR MOTORS >90kW)
- 49 63 71 — TRANSFORMER PROTECTIVE DEVICES (REFER ANSI STANDARDS FOR DESCRIPTIONS)
- CIRCUIT BREAKER (NOTE 11)
- xxxA — ST TRIP SET POINT; xxxA — CB SIZE
- AFD — ARC FLASH DETECTION
- ECR — EARTH CONTINUITY RELAY
- E-SHUT — SWITCHROOM EMERGENCY SHUTDOWN
- 63A 3 PHASE SOCKET OUTLET

DRG No.	REFERENCES	REV	DATE	DESCRIPTION	DRAW	CHECK	DESIGN	TECH APP	PROJ APP
2000-F13-SLD-EL-10411	F13-SW-114 415V ESSENTIAL SWITCHBOARD SINGLE LINE DIAGRAM								
2000-F13-SCM-EL-10442	F13-SW-114 415V ESSENTIAL SWITCHBOARD FEEDER SCHEMATIC & TERMINATION DIAGRAM								
2000-F13-SCM-EL-10403	F13-SW-014 415V MCC ARC FLASH SYSTEM SCHEMATIC DIAGRAM								
2000-F13-SCM-EL-10402	F13-SW-014 415V MCC 24VDC DISTRIBUTION SCHEMATIC DIAGRAM								
2000-F13-SCM-EL-10401	F13-SW-014 415V MCC INCOMER SCHEMATIC DIAGRAM								
2000-F13-DRG-EL-10106	F13-SR-004 415V SWITCHROOM ELECTRICAL EQUIPMENT LAYOUT								
2000-F13-BLK-IC-10401	F13-SR-004 415V SWITCHROOM AND REMOTE I/O COMMUNICATIONS BLOCK DIAGRAM								
2000-F13-SLD-EL-10402	F13-SW-014 415V MCC SINGLE LINE DIAGRAM SHEET 2	2	07SEP26	KCGM GROWTH PROJECT - SP1 AS-BUILT					
2000-F00-SLD-EL-10002	F00-SW-010 33kV SWITCHBOARD SINGLE LINE DIAGRAM SHEET 1 OF 5	1	09JUL25	FAT REDLINES INCORPORATED					
2000-F00-STD-EL-10001	GENERAL NOTES AND STANDARD DRAWINGS INDEX	0	11JUN24	APPROVED FOR CONSTRUCTION					

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PROJECT: NORTHERN STAR RESOURCES LTD
KCGM GROWTH PROJECT

TITLE: MILLING & CLASSIFICATION - RECYCLE CRUSHING
F13-SW-014 415V MCC
SINGLE LINE DIAGRAM SHEET 1

SHEET	SCALE	DRG No.	REVISION
A1	NONE	2000-F13-SLD-EL-10401	2

O:\CAD PROJECTS\235 NORTHERN STAR RESOURCES LTD\235177 KCGM GROWTH PROJECT\03 DELIVERABLES\EL\F13 - MILLING & CLASSIFICATION\2000-F13-SLD-EL-10401.DWG 08SEP26 9:59AM ROBINLAW